

Towards a Privacy-Preserving Approach for Seated Posture Monitoring Using CNN-Based mmWave Radar Sensing

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Abstract

Prolonged sedentary behavior has become increasingly common as more people spend long hours seated in front of computer screens, often leading to poor postures and musculoskeletal disorders. Hence, there is a growing demand for intelligent posture monitoring systems to detect and correct seated postures. While computer vision has excelled in human activity recognition (HAR), its reliance on high-resolution visual data raises privacy concerns. To address this, we propose mmPostureNet, a privacy-preserving solution that leverages millimeter-wave (mmWave) radar and convolutional neural networks (CNN) to monitor seated postures. We modify the mmWave-based Assistive Rehabilitation System (MARS) architecture to develop a lightweight seated posture classification model, suitable for CPU-only training and practical for real-time edge deployment. We collect a new dataset of seated poses and develop a point cloud preprocessing pipeline. Evaluation with Leave-One-Subject-Out (LOSO) cross-validation and unseen environments shows 88% accuracy, supporting mmPostureNet's potential for real-world, privacy-preserving posture monitoring.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*; • **Hardware** → Sensor devices and platforms.

Keywords

mmWave sensing; convolutional neural networks; human activity recognition

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1 Introduction

In today's digital age, sedentary behavior is increasingly common as more people spend much of their day seated at computers. A typical

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Figure 1: mmWave-based seated posture monitoring system deployed in a real-world setting, such as a single chip radar clipped to a laptop.

desk-based worker spends around 70% of their workday seated, accumulating 9-10 hours of sitting time per day when including time at home [16]. Whether using a laptop or desktop, this computer-centered lifestyle often leads to poor postures. Over time, improper postures can contribute to a variety of health issues, including back pain, neck strain, and even digestive issues [12]. According to the National Institutes of Health, roughly 80% of Americans experience lower back pain at some point in their lives, with poor posture recognized as a significant contributing factor [6]. These trends highlight an urgent need for intelligent systems capable of detecting and correcting seated postures to reduce long-term health risks.

Computer vision is a powerful tool for human activity recognition (HAR), using RGB cameras for tasks like pose estimation and motion tracking [2, 13, 14]. Despite computer vision's precise detection capabilities, camera-based systems raise privacy concerns, especially in personal settings, such as homes and offices. As a result, a more privacy-preserving sensing method is sought.

Millimeter-wave (mmWave) radar has recently gained attention as a promising solution for HAR that protects user privacy [7, 10, 15]. Unlike cameras, mmWave radar creates sparse point cloud representations of humans, keeping the subject's identity anonymous. It offers a compelling trade-off: resolution high enough for practical use, yet low enough to protect personal identity. When paired with deep learning, neural networks can learn spatial and temporal features from these point clouds, enabling complex HAR tasks.

While mmWave-based systems like mmWave-based Assistive Rehabilitation System (MARS) [1] have succeeded in full-body analysis, they fall short in seated posture monitoring. Upper-body-only tasks are more challenging due to the lack of lower-body context and subtler pose differences. These limitations call for specialized models that can effectively monitor seated postures from sparse point clouds.

We present **mmPostureNet**, a privacy-preserving approach for seated posture monitoring using mmWave radar and convolutional

neural networks (CNN). Building on MARS, mmPostureNet introduces architectural modifications that caters the model specifically for seated posture classification. To further support the adaptation, we collect a new dataset of seated poses and develop a point cloud preprocessing pipeline. mmPostureNet’s lightweight design allows training on CPU-only machines and enables deployment on edge devices for real-time performance. To assess generalizability, we evaluate mmPostureNet using Leave-One-Subject-Out (LOSO) cross-validation and testing in unseen environments, achieving an average accuracy of 88% across experiments. These results demonstrate the feasibility of deploying mmPostureNet in everyday devices, such as a laptop, for privacy-preserving posture monitoring.

2 Relevant Works

Computer vision has become a go-to method for HAR due to large-scale training datasets and powerful deep learning models [4, 5, 8]. OpenPose [2] and PoseNet [13] extract full-body skeletons from RGB images in real-time, enabling applications in healthcare and fitness. Pose estimation models and other vision-based methods [9] have been leveraged for seated posture analysis to help detect poor postures. While highly accurate, these models require constant visual recording, raising privacy concerns in personal settings.

As a privacy-preserving alternative, mmWave radar has emerged as a promising sensing modality for HAR [7, 10, 15]. While mmWave-based posture monitoring research is growing, most works focus on full-body analysis, and the few addressing seated postures often lack practicality or robustness. One approach uses traditional machine learning classifiers (e.g., SVM, KNN) and relies on manual feature extraction, requiring complex signal processing and domain expertise [11]. Another requires a specialized hardware setup and extensive calibration, limiting real-world deployment [17]. Some mmWave-based approaches use deep learning, but still lack the precision required for an effective solution [3].

MARS [1] is a CNN model excelling in joint-level pose estimation from radar point clouds, with a mean absolute error (MAE) of only 5 cm. MARS effectively extracts features, but it targets full body analysis, and falls short in analyzing upper-body-only poses for posture monitoring. Therefore, we present **mmPostureNet**, a lightweight CNN specialized for seated posture classification from radar point clouds. We support mmPostureNet with a new dataset of seated postures and a point cloud preprocessing pipeline.

3 Methodology

3.1 Data Collection

We collected a dataset of 9,522 point cloud samples from seven subjects sitting in four different postures — upright, slouched, left leaning, and right leaning — and aimed to compose a balanced dataset. These four positions are targeted, as slouching and asymmetric shoulder alignments are common risk factors for musculoskeletal disorders [6]. All samples in this dataset were collected in our lab room, as seen in Figure 2. A commodity 77 GHz mmWave radar equipped with three TI 3x4 radar chips was utilized, forming a 12x16 virtual MIMO array. The radar was positioned 60 cm away from the subject and 1 m above the ground at table height.

To establish ground truth labels, each subject was instructed to sit in front of the radar and hold a specific posture — upright, slouched,

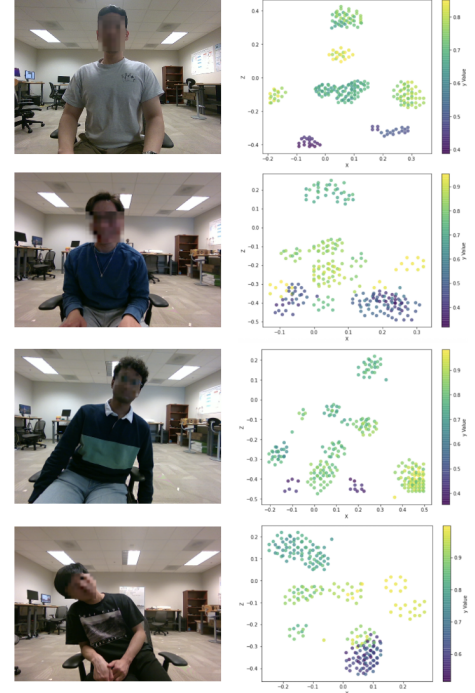


Figure 2: Preprocessed radar point clouds (viewed in 2D) with their corresponding seated postures in the lab room.

left leaning, or right leaning — for approximately five minutes per session. During these sessions, continuous point cloud data was captured and later segmented into individual samples. All samples from a session were assigned the corresponding posture label. The procedure was repeated for every posture across all subjects.

3.2 Preprocessing

The radar collects data as a 5D time series — 3D spatial coordinates (x, y, z), doppler, and intensity — collecting about 20 frames per second. For each dataset sample, we aggregate 10 frames to obtain a denser point cloud with about 1000-2000 points. Since mmPostureNet’s input size is 256 points, a preprocessing pipeline downsamples and filters the point clouds. First, any points lying outside the region of interest are discarded: $x \in [-60 \text{ cm}, 60 \text{ cm}]$, $y \leq 120 \text{ cm}$, $z \in [-60 \text{ cm}, 60 \text{ cm}]$ (x is horizontal axis, y is depth axis, z is vertical axis). Then, a voxel grid filter efficiently reduces points while still preserving spatial structure. DBSCAN clusters relevant points while removing isolated points, and any remaining points beyond 256 are removed using farthest point sampling, with the point cloud’s centroid as the starting point.

To prepare for model input, each point cloud sample is reshaped using a similar method as the original MARS work [1]. The points are first sorted by their x coordinate in ascending order. Any points with the same x coordinate are then sorted by their y coordinate, and if necessary, sorted by their z coordinate. This creates a 256-length vector with 5 features. The vector is then reshaped into a 16 x 16 square matrix in row-major order with shape (16, 16, 5), and can be interpreted as a 5-channelled 2D image.

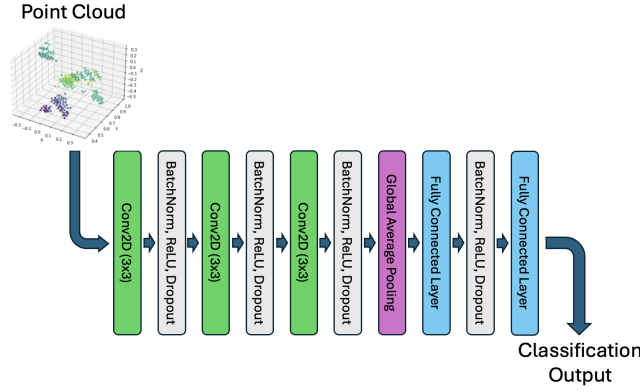


Figure 3: Lightweight CNN architecture of mmPostureNet

3.3 Model Architecture

The original MARS [1] includes two convolutional layers followed by flattening and fully connected layers. While effective, flattening a large spatial feature map increases the parameters in the fully connected layers, making the overall architecture heavier.

mmPostureNet is designed for improved efficiency and generalization. mmPostureNet includes an additional convolutional layer and replaces the flattening operation with a global average pooling layer. The added convolutional layer enables deeper feature extraction for subtle upper-body pose variations, while global average pooling produces a compact representation, reducing model complexity. The final layers consist of a fully connected layer and a classification layer, replacing MARS’s regression output layer.

Compared to MARS, mmPostureNet significantly reduces the number of learnable parameters, as seen in Table 1. A lighter model can potentially mitigate overfitting, especially with limited data, and improve generalization across different subjects and environments. Additionally, lighter models often lead to reduced latency.

Table 1: Parameter counts of MARS and mmPostureNet

MARS		mmPostureNet	
Layer (Config)	Params	Layer (Config)	Params
Conv2D (3×3, 5 in, 16 out)	736	Conv2D (3×3, 5 in, 16 out)	736
Conv2D (3×3, 16 in, 32 out)	4,640	Conv2D (3×3, 16 in, 32 out)	4,640
Flattening	0	Conv2D (3×3, 32 in, 64 out)	18,496
FC (2048 in, 512 out)	1,049,088	Global Avg Pooling	0
FC (512 in, 4 out)	2,052	FC (64 in, 128 out)	8,320
		FC (128 in, 4 out)	516
Total Params	1,056,516	Total Params	32,708

4 Evaluation

4.1 Training

LOSO cross-validation was performed to evaluate the robustness and generalizability of mmPostureNet. 7 training and testing iterations were conducted with each iteration using 6 subjects’ data for training and validation, and 1 unseen subject for testing. During each LOSO fold, data augmentation was applied exclusively to the training set to increase diversity. We added random jitter to x and z

coordinates and slightly scaled all spatial coordinates. The number of training samples doubled after augmentation. The training set was then normalized per sample to zero mean and unit variance. Validation and test sets were similarly normalized per sample, but without augmentation. All training iterations used a batch size of 128 and learning rate of 0.01 with early stopping enabled, and used a cross-entropy loss function. mmPostureNet was trained on a MacBook Pro with a 10-core M4 chip and 16 GB RAM.

Table 2: mmPostureNet LOSO cross-validation results

Subject	Accuracy
Subject 1	74.98%
Subject 2	84.54%
Subject 3	92.39%
Subject 4	93.24%
Subject 5	86.98%
Subject 6	70.79%
Subject 7	96.31%
Average Accuracy	85.60%

4.2 LOSO Cross-Validation Results

Table 2 summarizes the results of 7 iterations of LOSO cross-validation, with an average accuracy of 85.60% across all subjects. Figure 4 displays the aggregated confusion matrix across all 7 folds. While mmPostureNet performed well for most subjects, achieving over 90% accuracy in several cases, lower accuracies for certain individuals suggest opportunities to further improve generalization.

4.3 Unseen Environments Results

We trained mmPostureNet on the original dataset, which included all 7 subjects’ data collected in the lab room, and tested it in two new unseen environments without fine-tuning. We collected 1114 more samples in the conference room and 1260 samples in a small meeting room. The accuracies were 88.51% and 90.71%, respectively. Figure 4 displays the resulting confusion matrices.

5 Discussion

While our approach performs well overall, a few subjects scored notably lower accuracies. In these cases, the point clouds lacked clear clustered structures and remained scattered after preprocessing. Because mmWave radars are sensitive, small variations in body type, clothing, or sitting style can lead to dispersed point clouds. This challenges fixed-parameter filters like DBSCAN, which may remove valid points or retain noise when the distribution deviates from expected patterns. This issue is reflected in the confusion matrices, where misclassifications often default to the slouched class. Unlike upright or leaning poses, which have more defined directional structure, slouching can appear more flat and spread out. When preprocessing fails to impose structure, the model resorts to predicting the slouched class. For future work, we plan to enhance preprocessing with adaptive clustering that dynamically adjusts to various point cloud structures. Feature-guided sampling (e.g., based on point intensity) may also retain valuable points. In general, we

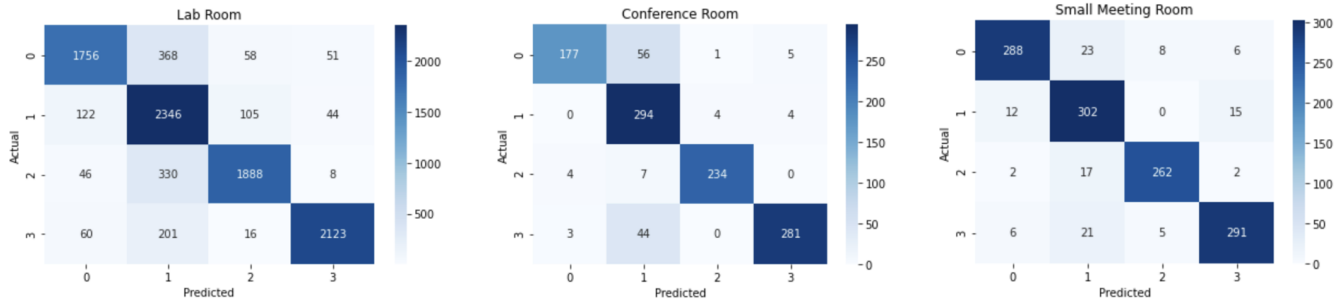


Figure 4: Confusion matrices for mmPostureNet's performance in three environments: lab room (left), conference room (middle), and small meeting room (right). Labels: 0 - upright, 1 - slouched, 2 - left leaning, 3 - right leaning.

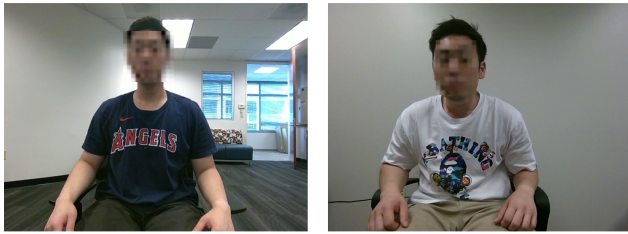


Figure 5: mmPostureNet tested in two unseen environments: conference room (left) and small meeting room (right).

aim to build a larger, more diverse dataset. Expanding the dataset may require a deeper network for more complex feature extraction, but we strive to keep a lightweight design for practical deployment. Finally, we aim to implement this system using a single-chip radar for integration in everyday devices, such as laptops.

6 Conclusion

In this work, we proposed **mmPostureNet**, a privacy-preserving seated posture monitoring system leveraging mmWave radar and a lightweight CNN. mmPostureNet classifies seated postures from sparse radar point clouds, achieving an average accuracy of 88% across unseen subjects and environments. Future work focuses on enhancing the preprocessing pipeline with adaptive clustering to better handle point clouds of varying spatial structures, expanding the dataset for greater subject diversity, and implementing the system with a single-chip radar for deployment on everyday devices.

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References

- [1] Sizhe An and Umit Y. Ogras. 2021. MARS: mmWave-based Assistive Rehabilitation System for Smart Healthcare. *ACM Trans. Embed. Comput. Syst.* 20, 5s, Article 72 (Sept. 2021), 22 pages. doi:10.1145/3477003
- [2] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. 2021. OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields. *IEEE Transactions on Pattern Analysis & Machine Intelligence* 43, 01 (Jan. 2021), 172–186. doi:10.1109/TPAMI.2019.2929257
- [3] Han Cui and Naim Dahnoun. 2022. Real-Time Short-Range Human Posture Estimation Using mmWave Radars and Neural Networks. *IEEE Sensors Journal* 22, 1 (2022), 535–543. doi:10.1109/JSEN.2021.3127937
- [4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*. 248–255. doi:10.1109/CVPR.2009.5206848
- [5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiuhua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=YicbFdNTTy>
- [6] Shu-Hao Du, Yong-Hui Zhang, Qi-Hao Yang, Yu-Chen Wang, Yu Fang, and Xue-Qiang Wang. 2023. Spinal posture assessment and low back pain. *EFORT Open Reviews* 8, 9 (September 2023), 708–718. doi:10.1530/EOR-23-0025
- [7] Tianbo Gu, Zheng Fang, Zhicheng Yang, Pengfei Hu, and Prasant Mohapatra. 2019. mmSense: Multi-Person Detection and Identification via mmWave Sensing. In *Proceedings of the 3rd ACM Workshop on Millimeter-Wave Networks and Sensing Systems* (Los Cabos, Mexico) (*mmNets '19*). Association for Computing Machinery, New York, NY, USA, 45–50. doi:10.1145/3349624.3356765
- [8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 770–778. doi:10.1109/CVPR.2016.90
- [9] Niklas Hoefflin, Tim Spulak, André Jeworutzki, and Jan Schwarzer. 2024. Real-Time Lateral Sitting Posture Detection using YOLOv5. In *2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechanics (BioRob)*. 711–715. doi:10.1109/BioRob60516.2024.10719953
- [10] Hao Kong, Xiangyu Xu, Jiadi Yu, Qilin Chen, Chenguang Ma, Yingying Chen, Yi-Chao Chen, and Linghe Kong. 2022. m3Track: <u>mm</u>-wave-based <u>multi-user 3D posture tracking</u>. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services* (Portland, Oregon) (*MobiSys '22*). Association for Computing Machinery, New York, NY, USA, 491–503. doi:10.1145/3498361.3538926
- [11] Guoxiang Liu, Xingguang Li, Chunsheng Xu, Lei Ma, and Hongye Li. 2023. FMCW Radar-Based Human Sitting Posture Detection. *IEEE Access* 11 (2023), 102746–102756. doi:10.1109/ACCESS.2023.3312328
- [12] Donald R. Murphy. 2024. Posture and How It Affects Your Health. <https://www.brownhealth.org/be-well/posture-and-how-it-affects-your-health>.
- [13] George Papandreou, Tyler Zhu, Nori Kanazawa, Alexander Toshev, Jonathan Tompson, Chris Bregler, and Kevin Murphy. 2017. Towards Accurate Multi-person Pose Estimation in the Wild. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 3711–3719. doi:10.1109/CVPR.2017.395
- [14] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. 2015. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. In *Advances in Neural Information Processing Systems*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett (Eds.), Vol. 28. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2015/file/14bfa6bb14875e45bba028a21ed38046-Paper.pdf
- [15] Arindam Sengupta, Feng Jin, Renyuan Zhang, and Siyang Cao. 2020. mm-Pose: Real-Time Human Skeletal Posture Estimation Using mmWave Radars and CNNs. *IEEE Sensors Journal* 20, 17 (2020), 10032–10044. doi:10.1109/JSEN.2020.2991741
- [16] University Hospitals of Leicester NHS Trust. 2022. Reducing sitting time to boost health in office workers. <https://medicalxpress.com/news/2022-08-boost-health-office-workers.html>.
- [17] Bo Zhang, Zimeng Zhou, Boyu Jiang, and Rong Zheng. 2024. SUPER: Seated Upper Body Pose Estimation using mmWave Radars. In *2024 IEEE/ACM Ninth International Conference on Internet-of-Things Design and Implementation (IoTDI)*. IEEE, 181–191. doi:10.1109/iotdi61053.2024.00020